

Challenge (Habanero)

- Q1.** A hiking trail has a slope of 2. What is the slope of a perpendicular trail?
(A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) 2
(D) -2
(E) 0
- Q2.** If a line has a slope of -3 and passes through (1, 2), what is the equation of a perpendicular line passing through (1, 2)?
(A) $y = \frac{1}{3}x + \frac{5}{3}$
(B) $y = -\frac{1}{3}x + \frac{5}{3}$
(C) $y = \frac{1}{3}x - \frac{5}{3}$
(D) $y = -\frac{1}{3}x - \frac{5}{3}$
(E) $y = \frac{1}{3}x + 1$
- Q3.** A fuel consumption study found that the rate of fuel consumption is -2 gallons per mile. What is the slope of a line perpendicular to this rate?
(A) $\frac{1}{2}$
(B) $-\frac{1}{2}$
(C) 2
(D) -2
(E) 0
- Q4.** A animal population study modeled the population growth with the equation $y = -4x + 10$. What is the equation of a line perpendicular to this model?
(A) $y = \frac{1}{4}x + 10$
(B) $y = -\frac{1}{4}x + 10$
(C) $y = \frac{1}{4}x - 10$
(D) $y = -\frac{1}{4}x - 10$
- (E) $y = \frac{1}{4}x + 5$
- Q5.** A line has a slope of $-\frac{3}{4}$ and passes through (2, 5). What is the slope of a line perpendicular to this line?
(A) $\frac{4}{3}$
(B) $-\frac{4}{3}$
(C) $\frac{3}{4}$
(D) $-\frac{3}{4}$
(E) 0
- Q6.** If two lines are perpendicular, and one line has a slope of 5, what is the slope of the other line?
(A) $-\frac{1}{5}$
(B) $\frac{1}{5}$
(C) 5
(D) -5
(E) 0
- Q7.** A line passes through the points (1, 2) and (3, 4). What is the slope of a line perpendicular to this line?
(A) -1
(B) 1
(C) 2
(D) -2
(E) 0
- Q8.** The equation of a line is $y = 2x - 5$. What is the equation of a line perpendicular to this line and passing through (0, 0)?
(A) $y = -\frac{1}{2}x$
(B) $y = \frac{1}{2}x$
(C) $y = -2x$
(D) $y = 2x$
(E) $y = -\frac{1}{2}x + 1$

Open Ended Questions (FRQ)

- Q9.** A park ranger is designing a new hiking trail. The trail will have a slope of -2 and will pass through the point (3, 4). Find the equation of a line that is perpendicular to the trail and passes through the point (3, 4). Show all work and explain your reasoning.
- Q10.** A wildlife biologist is studying the population of a certain species of animal. The population is modeled by the equation $y = -3x + 12$. Find the equation of a line that is perpendicular to this model and passes through the point (2, 6). Show all work and explain your reasoning.

Chef's Tips

- Make sure to read each question carefully and understand what is being asked.
- Use the negative reciprocal to find the slope of a perpendicular line.
- Show all work and explain your reasoning for free response questions.